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Wedge-lock sheath retention mechanism

Abstract

A medical device system may include a sheath, a pusher wire, and a locking element. The pusher wire may be slidably disposed within a lumen of the sheath. The locking element may have a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end. The locking element may have a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first outer diameter. The distal end region of the locking element may be configured to freely slide over the sheath and when a proximal end region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority of U.S. Provisional Application No. 62/904,864 filed Sep. 24, 2019, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD
(1) The present disclosure pertains to medical devices and methods for manufacturing and/or using medical devices. More particularly, the present disclosure pertains to configurations of a system for securing a medical device within a sheath.
BACKGROUND
(2) A wide variety of intracorporeal medical devices have been developed for medical use, for example, surgical and/or intravascular use. Some of these devices include guidewires, catheters, medical device delivery systems (e.g., for stents, grafts, replacement valves, etc.), and the like. These devices are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. There is an ongoing need to provide alternative medical devices as well as alternative methods for manufacturing and/or using medical devices.
SUMMARY
(3) In a first example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end. The locking element may have a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first inner diameter. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the locking element is disposed over

the sheath, the locking element may be configured to depress the sheath radially inwards.

(4) Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.

(5) Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the distal end region may have a generally uniform inner diameter.

(6) Alternatively or additionally to any of the examples above, in another example, the sheath may include a first slot and a second slot, the first and second slots extending distally from the proximal end of the sheath and extending less than an entire length of the sheath.

(7) Alternatively or additionally to any of the examples above, in another example, the sheath may include a longitudinally extending slot extending distally from the proximal end of the sheath.

(8) Alternatively or additionally to any of the examples above, in another example, the longitudinally extending slot may terminate at a circumferentially extending slit.

(9) Alternatively or additionally to any of the examples above, in another example, the sheath may include an angled protrusion extending radially outward from an outer surface thereof.

(10) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may reduce in diameter in a proximal direction.

(11) Alternatively or additionally to any of the examples above, in another example, the locking element may include an angled protrusion extending radially inward from an inner surface of the lumen of the locking element.

(12) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may reduce in diameter in a distal direction.

(13) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath and the angled protrusion of the locking element may be configured to cooperate to limit axial movement of the locking element.

(14) Alternatively or additionally to any of the examples above, in another example, the locking element may further comprise a slot extending from a proximal end of the locking element to a distal end of the locking element, the slot extending through a thickness of a wall of the locking element.

(15) Alternatively or additionally to any of the examples above, in another example, the slot may have a width that is greater than a width of the pusher wire and less than a width of the sheath.

(16) Alternatively or additionally to any of the examples above, in another example, the locking element may comprise a plurality of circumferentially extending ribs extending from an outer surface thereof.

(17) Alternatively or additionally to any of the examples above, in another example, the locking element may be formed from a more rigid material than the sheath.

(18) In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath having a first slot and a second slot, the first and second slots extending distally from the proximal end of the sheath, extending less than an entire length of the sheath, and defining a first flexible arm and a second flexible arm, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the locking element having a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first inner diameter. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the locking element is disposed over a proximal end region of the sheath, the locking element may be configured to depress the first and second flexible arms radially inwards.

(19) Alternatively or additionally to any of the examples above, in another example, the lumen of

the locking element within proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.
(20) Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the distal end region may have a generally uniform inner diameter.
(21) Alternatively or additionally to any of the examples above, in another example, the locking element may further comprise a slot extending from a proximal end of the locking element to a distal end of the locking element, the slot extending through a thickness of a wall of the locking element.
(22) Alternatively or additionally to any of the examples above, in another example, the slot may have a width that is greater than a width of the pusher wire and less than a width of the sheath.
(23) Alternatively or additionally to any of the examples above, in another example, the locking element may comprise a plurality of circumferentially extending ribs extending from an outer surface thereof.
(24) Alternatively or additionally to any of the examples above, in another example, the locking element may have a generally uniform wall thickness from the proximal end to the distal end of the locking element.
(25) Alternatively or additionally to any of the examples above, in another example, the locking element may have a first wall thickness adjacent the proximal end and a second wall thickness adjacent to the distal end, the first wall thickness greater than the second wall thickness.
(26) In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the lumen of the locking element having a generally hourglass shape. A distal end region of the locking element may be configured to freely slide over the sheath. When an intermediate region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.
(27) Alternatively or additionally to any of the examples above, in another example, the longitudinally extending slot of the sheath may terminate at a circumferentially extending slit.
(28) Alternatively or additionally to any of the examples above, in another example, the sheath may further comprise a first flap and a second flap adjacent the longitudinally extending slot.
(29) Alternatively or additionally to any of the examples above, in another example, the locking element may have a first inner diameter adjacent the distal end and a second inner diameter adjacent to the intermediate region, the second inner diameter smaller than the first inner diameter.
(30) Alternatively or additionally to any of the examples above, in another example, the second inner diameter may be generally constant and extends between a flared distal end region and a flared proximal end region.
(31) In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising an angled protrusion extending radially outward from an outer surface thereof, a pusher wire slidably disposed within the lumen of the sheath, and a locking element. The locking element may comprise a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, a first inner diameter adjacent to the distal end, a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first outer diameter, and an angled protrusion extending radially inward from an inner surface of the lumen of the locking element. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the

locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.
(32) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may reduce in diameter in a proximal direction.
(33) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may be positioned adjacent to the proximal end of the sheath.
(34) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may reduce in diameter in a distal direction.
(35) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may be positioned adjacent to the distal end of the locking element.
(36) Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath and the angled protrusion of the locking element may be configured to cooperate to limit axial movement of the locking element.
(37) Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.
(38) The above summary of some embodiments, aspects, and/or examples is not intended to describe each embodiment or every implementation of the present disclosure. The figures and the detailed description which follows more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:
(2) FIG. **1** is a perspective view of an example medical device system in a first configuration;
(3) FIG. **2** is a perspective view of an illustrative locking mechanism;
(4) FIG. **3** is a cross-sectional view of the illustrative locking mechanism of FIG. **2**;
(5) FIG. **4** is a perspective view of another illustrative locking mechanism;
(6) FIG. **5** is a cross-sectional view of the illustrative locking mechanism of FIG. **4**;
(7) FIG. **6** is a perspective view of another illustrative locking mechanism;
(8) FIG. **7** is a cross-sectional view of the illustrative locking mechanism of FIG. **6**;
(9) FIG. **8** is a perspective view of the illustrative medical device system of FIG. **1** in a second configuration;
(10) FIG. **9** is a partial cross-sectional view of the illustrative medical device system of FIG. **8**;
(11) FIG. **10** is a perspective view of the illustrative medical device system of FIGS. **1** and **8** in a third configuration;
(12) FIG. **11** is a partial cross-sectional view of the illustrative medical device system of FIG. **10**;
(13) FIG. **12** is a perspective view of another example medical device system in a first configuration;
(14) FIG. **13** is a partial cross-sectional view of the illustrative medical device system of FIG. **12**;
(15) FIG. **14** is a perspective view of another illustrative locking mechanism;
(16) FIG. **15** is a cross-sectional view of the illustrative locking mechanism of FIG. **14**;
(17) FIG. **16** is a perspective view of the illustrative medical device system of FIG. **12** in a second configuration;
(18) FIG. **17**A is a partial cross-sectional view of the illustrative medical device system of FIG. **16**;
(19) FIG. **17**B is a partial cross-sectional view of the illustrative medical device system of FIG. **17**A;

(20) FIG. **18** is a perspective view of another example medical device system in a first configuration;
(21) FIG. **19** is a partial cross-sectional view of the illustrative medical device system of FIG. **18**;
(22) FIG. **20** is a partial cross-sectional view of the illustrative medical device system of FIG. **18** in a second configuration;
(23) FIG. **21** is a perspective view of another illustrative locking mechanism;
(24) FIG. **22** is a cross-sectional view of the illustrative locking mechanism of FIG. **21**;
(25) FIG. **23** is a perspective view of another example medical device system in a first configuration;
(26) FIG. **24** is a perspective view of an illustrative locking mechanism;
(27) FIG. **25** is a cross-sectional view of the illustrative locking mechanism of FIG. **24**;
(28) FIG. **26** is a distal end view of the illustrative locking mechanism of FIG. **24**;
(29) FIG. **27** is a perspective view of an illustrative locking mechanism;
(30) FIG. **28** is a perspective cross-sectional view of the illustrative locking mechanism of FIG. **27**; and
(31) FIG. **29** is a distal end view of the illustrative locking mechanism of FIG. **27**.
(32) While aspects of the disclosure are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

(33) The following description should be read with reference to the drawings, which are not necessarily to scale, wherein like reference numerals indicate like elements throughout the several views. The detailed description and drawings are intended to illustrate but not limit the claimed invention. Those skilled in the art will recognize that the various elements described and/or shown may be arranged in various combinations and configurations without departing from the scope of the disclosure. The detailed description and drawings illustrate example embodiments of the claimed invention. However, in the interest of clarity and ease of understanding, while every feature and/or element may not be shown in each drawing, the feature(s) and/or element(s) may be understood to be present regardless, unless otherwise specified.
(34) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
(35) All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about”, in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (e.g., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.
(36) The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
(37) Although some suitable dimensions, ranges, and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges, and/or values may deviate from those expressly disclosed.
(38) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise. It is to be noted that in order to facilitate understanding,

certain features of the disclosure may be described in the singular, even though those features may be plural or recurring within the disclosed embodiment(s). Each instance of the features may include and/or be encompassed by the singular disclosure(s), unless expressly stated to the contrary. For simplicity and clarity purposes, not all elements of the disclosed invention are necessarily shown in each figure or discussed in detail below. However, it will be understood that the following discussion may apply equally to any and/or all of the components for which there are more than one, unless explicitly stated to the contrary. Additionally, not all instances of some elements or features may be shown in each figure for clarity.

(39) Relative terms such as “proximal”, “distal”, “advance”, “retract”, variants thereof, and the like, may be generally considered with respect to the positioning, direction, and/or operation of various elements relative to a user/operator/manipulator of the device, wherein “proximal” and “retract” indicate or refer to closer to or toward the user and “distal” and “advance” indicate or refer to farther from or away from the user. In some instances, the terms “proximal” and “distal” may be arbitrarily assigned in an effort to facilitate understanding of the disclosure, and such instances will be readily apparent to the skilled artisan. Other relative terms, such as “upstream”, “downstream”, “inflow”, and “outflow” refer to a direction of fluid flow within a lumen, such as a body lumen, a blood vessel, or within a device. Still other relative terms, such as “axial”, “circumferential”, “longitudinal”, “lateral”, “radial”, etc. and/or variants thereof generally refer to direction and/or orientation relative to a central longitudinal axis of the disclosed structure or device.

(40) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect the particular feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

(41) For the purpose of clarity, certain identifying numerical nomenclature (e.g., first, second, third, fourth, etc.) may be used throughout the description and/or claims to name and/or differentiate between various described and/or claimed features. It is to be understood that the numerical nomenclature is not intended to be limiting and is exemplary only. In some embodiments, alterations of and deviations from previously-used numerical nomenclature may be made in the interest of brevity and clarity. That is, a feature identified as a “first” element may later be referred to as a “second” element, a “third” element, etc. or may be omitted entirely, and/or a different feature may be referred to as the “first” element. The meaning and/or designation in each instance will be apparent to the skilled practitioner.

(42) Diseases and/or medical conditions that impact and/or are affected by the cardiovascular system are prevalent throughout the world. For example, some forms of arterial venous malformations (AVMs) may “feed” off of normal blood flow through the vascular system. Without being bound by theory, it is believed that it may be possible to treat, at least partially, arterial venous malformations and/or other diseases or conditions by starving them of normal, oxygen and/or nutrient-rich blood flow, thereby limiting their ability to grow and/or spread. Other examples of diseases or conditions that may benefit from vascular occlusion include, but are not limited to, bleeds, aneurysms, venous insufficiency, shutting off blood flow prior to organ resection, or preventing embolic bead reflux into branch vessels in the liver. Disclosed herein are

medical devices that may be used within a portion of the cardiovascular system in order to treat and/or repair some arterial venous malformations and/or other diseases or conditions. The devices disclosed herein may also provide a number of additional desirable features and benefits as described in more detail below.
(43) FIG. **1** is a perspective view of an example medical device system **100** in a partially unassembled configuration. The medical device system **100** may include a pusher wire **102**, an implant **104** (see, for example, FIG. **9**), such as, but not limited to, an embolic coil, an introducer sheath **106**, and a locking element **108**. For simplicity, the implant **104** is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.
(44) Embolic coils **104** may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **106** containing the coil **104** can be used to carry and protect the coil **104** prior to insertion into the patient. Further, the introducer sheath **106** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **106** may be configured to protect the implant **104** and maintain the implant **104** in a deliverable orientation, until the implant **104** is deployed. As will be described in more detail herein, the locking element **108** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **102** and implant **104** within the sheath **106** until the user is ready to advance the implant **104** out of the sheath **106**.
(45) The sheath **106** may be a tubular member including a proximal end **112**, a distal end **114**, and an intermediate region **116** positioned therebetween. Some suitable but non-limiting materials for the sheath **106**, for example, polymer materials, composite materials, etc., are described below. The sheath **106** may define a lumen **110** extending from the proximal end **112** to the distal end **114**. The pusher wire **102** and implant **104** may be slidably disposed within the lumen **110** of the sheath **106** such that the pusher wire **102** and the implant **104** are radially inwards of the sheath **106**. The implant **104** may be disposed proximate the distal end **114** of the sheath **106**. The pusher wire **102** may be axially slidable between an interlocked position and a released position. The pusher wire **102** may be configured to be releasably attached to the implant **104**. The implant **104** may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **102** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **102**, for example, metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **102** may be releasably secured to the sheath **106** via the locking element **108** to limit axial and/or rotational movement of the pusher wire **102** within the sheath **106**.
(46) The sheath **106** may have a first slot **118***a* and a second slot **118***b* (collectively, **118**) extending distally from the proximal end **112**. The first and second slots **118***a*, **118***b* may be positioned across from another or spaced approximately 180° about the circumference of the sheath **106**. While the sheath **106** is described as having two slots, the sheath **106** may include fewer than two or more than two slots, as desired. The slots **118** may extend less than an entire length of the sheath **106**. The slots **118** may remove material from the sheath **106** to create flexible arms or members **120***a*, **120***b* (collectively, **120**). While the sheath **106** is described as having two arms **120**, it should be understood the number of flexible arms may vary with the number of slots **118** and there can be fewer than two or more than two flexible arms, as desired. Further, the slots **118** may be uniformly or eccentrically distributed about a circumference of the sheath **106**. The length and/or size of the slots **118** (and/or the arms **120**) may vary to produce different degrees of wedging (between the locking element **108** and the pusher wire **102**) and locking capabilities. It is contemplated that a

proximal end region **129** of the sheath **106** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **129** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **106** may be formed from a same material as the proximal end region **129** while in other cases, other portions of the sheath **106** may be formed from a different material as the proximal end region **129**. For example, the sheath **106** may include a polyimide distal tip (although this is not required).
(47) Referring additionally to FIG. **2**, which illustrates a perspective view of the locking element **108**, the locking element **108** may be a tubular member having a proximal end **122**, a distal end **124**, and an intermediate region **126** positioned therebetween. The locking element **108** may have a generally constant or uniform outer diameter **130** over a proximal region **128** extending from the proximal end **122** towards and into the intermediate region **126**. The locking element **108** may include a distal region **132** including plurality of raised regions or circumferential ribs **136***a*, **136***b*, **136***c*, **136***d* (collectively, **136**). The circumferential ribs **136** may have an outer diameter **134** that is greater than the outer diameter **130** of the proximal region **128**. It is contemplated that the circumferential ribs **136** may increase the tactility of the locking element **108** and/or make the locking element **108** easier to grip or handle. The number of circumferential ribs **136** (e.g., fewer than four or greater than four), the size of the circumferential ribs **136** (e.g., increasing or decreasing the diameter **134**), the geometry of the circumferential ribs **136** (extending about less than an entire circumference, having a different shape, etc.), positioning of the circumferential ribs **136** along a length of the locking element **108**, and/or the spacing of the circumferential ribs **136** can be varied to enhance or diminish the tactility or gripability of the locking element **108**, as desired.
(48) The locking element **108** may define a lumen **135** extending from the proximal end **122** to the distal end **124**. As will be discussed in more detail herein, a portion of the lumen **135** of the locking element **108** may be sized to slide freely over the sheath **106** while another portion of the lumen **135** may be sized to exert a radially inward compressing force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **108**.
(49) The locking element **108** may be formed in a variety of different manners. For example, the locking element **108** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **108** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.
(50) FIG. **3** is a cross-sectional view of the locking element **108** taken at line **3-3** of FIG. **2**. The locking element **108** may have a variable inner diameter. For example, in some embodiments, the locking element **108** may have an inner diameter which increases in a sloped manner over a first length **140** in the distal direction from a first inner diameter **137** adjacent the proximal end **122** to a second inner diameter **138**. In some embodiments, the first length **140** may be approximately the same as a length of the arms **120** of the sheath **106**, although this is not required. The second inner diameter **138** may be approximately constant or uniform over a second length **142** of the locking element **108**. In some cases, while not explicitly shown, the transition from the first inner diameter **137** to the second inner diameter **138** may be an abrupt or step-wise transition. The first length **140** and/or the second length **142** may vary, as desired. Further, the slope of the first length **140** may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered first length **140** and/or the second length **142** may be changed to create the desired locking effect (described in more detail herein).
(51) In some embodiments, a wall thickness of the locking element **108** may vary inversely to the inner diameter over the first length **140** while the outer diameter **130** remains constant to vary the inner diameter. For example, the locking element **108** may have a wall thickness **144** adjacent the proximal end **122**. The wall thickness may gradually decrease to a second wall thickness **146** at or near the second length **142**. In other embodiments, the wall thickness may remain constant. In such

an instance, the inner diameter and the outer diameter of the locking element **108** may vary (e.g., may be sloped in a similar manner) along the first length **140**. For example, the outer diameter of the locking element **108** may increase from the proximal end **122** over the first length **140**, as the inner diameter increases.

(52) The second, larger inner diameter **138** may be sized such that the locking element **108** can slide freely over the sheath **106**. The first, smaller inner diameter **137** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **106**, as will be described in more detail herein.

(53) It is contemplated that the configuration of the locking element **108** may be adjusted to create the desired effect. For example, one or more of the inner diameters **137**, **138** may be made larger or smaller to accommodate different sizes of sheaths **106** and/or different outer diameters of the pusher wire **102**. Further, the first and/or second lengths **140**, **142** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **108** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **108** may include features in addition to, or in place of the circumferential ribs **136** to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **108** may include visual indicia to guide the user in manipulation of the locking element **108**. In some cases, the overall length of the locking element **108** can be increased or decreased, as desired. It is further contemplated that the locking element **108** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to, packaging.

(54) FIG. **4** illustrates a perspective view of another illustrative locking element **200**. The locking element **200** may be similar in form and function to the illustrative locking mechanism **108** described herein and may be used with a medical device system, such as the medical device system **100** described herein. The locking element **200** may be a tubular member having a proximal end **202**, a distal end **204**, and an intermediate region **206** positioned therebetween. The locking element **200** may have a generally constant or uniform outer diameter **210** extending from the proximal end **202** to the distal end **204**, although this is not required. The locking element **200** may define a lumen **212** extending from the proximal end **202** to the distal end **204**. As will be discussed in more detail herein, a portion of the lumen **212** of the locking element **200** may be sized to slide freely over a sheath, such as the sheath **106** described herein, while another portion of the lumen **212** may be sized to exert a radially inward or compressive force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **200**.

(55) The locking element **200** may be formed in a variety of different manners. For example, the locking element **200** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **200** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

(56) FIG. **5** is a cross-sectional view of the locking element **200** taken at line **5-5** of FIG. **4**. The locking element **200** may have a variable inner diameter. For example, in some embodiments, the locking element **200** may have an inner diameter which increases in a sloped manner over a first length **214** in the distal direction from a first inner diameter **216** adjacent the proximal end **202** to a second inner diameter **218**. In some embodiments, the first length **214** may be approximately the same as a length of the arms **120** of the sheath **106**, although this is not required. The second inner diameter **218** may be approximately constant or uniform over a second length **220** of the locking element **200**. In some cases, while not explicitly shown, the transition from the first inner diameter **216** to the second inner diameter **218** may be an abrupt or step-wise transition. The first length **214** and/or the second length **220** may vary, as desired. Further, the slope of the first length **214** may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the

tapered first length **214** and/or the second length **220** may be changed to create the desired locking effect (described in more detail herein).

(57) In some embodiments, a wall thickness of the locking element **200** may vary inversely to the inner diameter over the first length **214** while the outer diameter **210** remains constant to vary the inner diameter. For example, the locking element **200** may have a first wall thickness **222** adjacent the proximal end **202**. The wall thickness may gradually decrease to a second wall thickness **224** at or near the second length **220**. In other embodiments, the wall thickness may remain constant. In such an instance, the inner diameter and the outer diameter of the locking element **200** may vary (e.g., may be sloped in a similar manner) along the first length **214**. For example, the outer diameter of the locking element **200** may increase from the proximal end **202** over the first length **214**, as the inner diameter increases.

(58) The second, larger inner diameter **218** may be sized such that the locking element **200** can slide freely over the sheath **106**. The first, smaller inner diameter **216** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **106**, as will be described in more detail herein.

(59) It is contemplated that the configuration of the locking element **200** may be adjusted to create the desired effect. For example, one or more of the inner diameters **216**, **218** may be made larger or smaller to accommodate different sizes of sheaths **106**. Further, the first and/or second lengths **214**, **220** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **200** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **200** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **200** may include visual indicia to guide the user in manipulation of the locking element **200**. In some cases, the overall length of the locking element **200** can be increased or decreased, as desired. It is further contemplated that the locking element **200** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(60) FIG. **6** illustrates a perspective view of another illustrative locking element **300**. The locking element **300** may be similar in form and function to the illustrative locking mechanism **108** described herein and may be used with a medical device system, such as the medical device system **100** described herein. The locking element **300** may be a tubular member having a proximal end **302**, a distal end **304**, and an intermediate region **305** positioned therebetween. The locking element **300** may define a lumen **312** extending from the proximal end **302** to the distal end **304**. As will be discussed in more detail herein, a portion of the lumen **312** of the locking element **300** may be sized to slide freely over a sheath, such as the sheath **106** described herein, while another portion of the lumen **312** may be sized to exert a radially inward or compressive force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **300**. The locking element **300** may further include a slot **310** extending from the proximal end **302** to the distal end **304**. The slot **310** may extend through a thickness of the side wall of the locking element **300** (e.g., from an outer surface to an inner surface thereof). The slot **310** may have a width **314** that is greater than a width of a pusher wire (e.g., such as the pusher wire **102** described herein) but less than a width of the sheath **106**. This may allow the locking element **300** to be more easily removed from the medical device system **100** as will be described in more detail herein.

(61) The locking element **300** may have a tapered or sloped outer diameter over a proximal region **328** extending from the proximal end **302** towards and into the intermediate region **305**. The locking element **300** may include a distal region **330** including plurality of raised regions or circumferential ribs **332***a*, **332***b*, **332***c*, **332***d* (collectively, **332**). The circumferential ribs **332** may have an outer diameter **334** that is greater than an outer diameter **324**, **326** of the proximal region **328**. The circumferential ribs **332** may extend about an entire perimeter of the locking element **300**,

which may be less than 360° due to the slot **310**. It is contemplated that the circumferential ribs **332** may increase the tactility of the locking element **300** and/or make the locking element **300** easier to grip or handle. The number of circumferential ribs **332** (e.g., fewer than four or greater than four), the size of the circumferential ribs **332** (e.g., increasing or decreasing the diameter **334**), the geometry of the circumferential ribs **332** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **332** can be varied to enhance or diminish the tactility or gripability of the locking element **300**, as desired.

(62) The locking element **300** may be formed in a variety of different manners. For example, the locking element **300** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **300** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

(63) FIG. **7** is a cross-sectional view of the locking element **300** taken at line **7-7** of FIG. **6**. The locking element **300** may have a variable outer diameter and a variable inner diameter. For example, in some embodiments, the locking element **300** may have an inner diameter which increases in a sloped manner over a first length **316** in the distal direction from a first inner diameter **318** adjacent the proximal end **302** to a second inner diameter **320**. In some embodiments, the first length **316** may be approximately the same as a length of the arms **120** of the sheath **106**, although this is not required. The second inner diameter **320** may be approximately constant or uniform over a second length **322** of the locking element **300**. In some cases, while not explicitly shown, the transition from the first inner diameter **318** to the second inner diameter **320** may be an abrupt or step-wise transition. The first length **316** and/or the second length **322** may vary, as desired. Further, the slope of the first length **316** may vary, as desired. The second, larger inner diameter **320** may be sized such that the locking element **300** can slide freely over the sheath **106**. The first, smaller inner diameter **318** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **106**, as will be described in more detail herein.

(64) In some embodiments, a wall thickness **336** of the locking element **300** may remain generally uniform over the first and/or second lengths **316**, **322** of the locking element **300**. In such an instance, the inner diameter and the outer diameter of the locking element **300** may vary (e.g., may be sloped in a similar manner) along the first length **316**. For example, the outer diameter of the locking element **300** may increase from a first outer diameter **324** at the proximal end **302** to a second outer diameter **326** (greater than the first outer diameter **324**) over the first length **316**, as the inner diameter increases. In some embodiments, the wall thickness **336** may vary over the first and/or second lengths **316**, **322** of the locking element **300**. For example, when circumferential ribs **332** are provided, the wall thickness adjacent the ribs **332** may be greater than the wall thickness **336** of other portions of the locking element **300**. In other embodiments, the wall thickness **336** may vary to vary the diameter of the lumen **310**.

(65) It is contemplated that the configuration of the locking element **300** may be adjusted to create the desired effect. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered first length **316** and/or the second length **322** may be changed to create the desired locking effect (described in more detail herein). For example, one or more of the inner diameters **318**, **320** may be made larger or smaller to accommodate different sizes of sheaths **106**. Further, the first and/or second lengths **316**, **322** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **300** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **300** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **300** may include visual indicia to guide the user in manipulation of the locking element **300**. In some cases, the overall length of the locking element **300** can be increased or decreased, as desired. It is further contemplated that the locking element **300** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(66) FIG. **8** is a perspective view of the illustrative medical device system **100** in an unlocked or first assembled configuration and FIG. **9** is a partial cross-sectional view of the illustrative medical device system **100**, taken at line **9-9** of FIG. **8**. While the medical device system **100** is described with respect to a particular locking element **108**, it should be understood that any of the locking elements described herein, including but not limited to, the locking elements **200**, **300**, **408**, **508**, **600**, **708**, **800**, may be substituted for the locking element **108**. From the partially unassembled configuration illustrated in FIG. **1**, to position the locking element **108** such that it can be used to secure the pusher wire **102**, the locking element **108** is distally advanced over a proximal end region **129** of the sheath **106** from the proximal end **112**. An entirety of the locking element **108** may not be disposed over sheath **106** in the first assembled configuration. For example, the first length **140** of the locking element **108** having the reduced inner diameter may be only partially disposed over the sheath **106** or not over the sheath **106** at all. In this configuration, the pusher wire **102** is free to slide axially (e.g., proximally and distally) and/or rotate within the lumen **110** of the sheath **106**. As described above, the inner diameter of the sheath **106** may remain constant, or substantially constant from the proximal end **112** to the distal end **114**.
(67) To lock the pusher wire **102** relative to the sheath **106**, the locking element **108** is advanced further in the distal direction until the first length **140** of the locking element **108** is disposed over the arms **120** of the sheath **106**. FIG. **10** is a perspective view of the illustrative medical device system **100** in a locked or second assembled configuration and FIG. **11** is a partial cross-sectional view of the illustrative medical device system **100**, taken at line **11-11** of FIG. **10**. As described above, the distal portion **132** and a portion of the proximal portion **128** of locking element **108** may slide freely over the proximal end region **129** of the sheath. When proximal portion **128** having the reduced diameter of the locking element **108** reaches the outer sheath **106**, the arms **120** of the sheath **106** may be depressed onto an outer surface of the pusher wire **102**. For example, the locking element **108** may freely slide over the outer sheath **106** until the inner diameter of the locking element **108** is approximately equal to or less than an outer diameter of the sheath **106**. It is contemplated that the wall thickness of the outer sheath **106** and the outer diameter of the pusher wire **102** may collectively affect how much the sheath arms **120** collapse before locking on the pusher wire **102**. Once the inner diameter of the locking element **108** begins to frictionally engage the outer surface of the sheath **106** more force may be required to continue distal movement of the locking element **108**. As the locking element **108** is distally advanced further onto the sheath **106**, the tapered region (e.g., the first length **140**) acts to deflect the arms **120** of the sheath **106** inward and onto the wire **102**. When advanced far enough, the locking element **108** wedges in place, creating the desired locking effect between the sheath **106** and the pusher wire **102**. For example, the tapered portion of the lumen **135** of the locking element **108** may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the arms **120** of the sheath **106** such that the outer diameter of the arms **120** is decreased and the inner diameter of the arms is biased radially inward, as shown in FIG. **11**. As the inner diameter of the arms **120** is reduced, the inner surface of the arms **120** contacts and frictionally engages an outer surface of the pusher wire **102**.
(68) In the locked configuration illustrated in FIGS. **10** and **11**, the frictional engagement between the inner surface of the sheath **106** and the outer surface of the pusher wire **102** may preclude or inhibit axial (e.g., proximally and distally) and/or rotational movement of the pusher wire **102** within the lumen **110** of the sheath **106**. In some embodiments, the arms **120** are configured to promote inward deflection thereof. For example, the cut pattern of the slots **118** may be applied to promote collapse of the arms **120** and to create a localized friction point between an inner surface of the sheath **106** (e.g., the inner surface of the arms **120**) and the pusher wire **102**.
(69) It is contemplated that the gradual decrease in the inner diameter of the locking element **108** may facilitate positioning of the locking element **108** over the sheath **106**. For example, the inner diameter **138** of the locking element **108** at the distal portion **132** (and/or portions of the proximal portion **128** may be large enough to freely pass over the outer diameter of the sheath **106**. As the

inner diameter of the locking element tapers (e.g., in the proximal direction towards the proximal end **122**) to a dimension smaller than the outer diameter of the sheath **106**, and is distally advanced far enough over the sheath **106**, it promotes collapse of the arms **120**. This creates a friction lock between the sheath **106** and pusher wire **102** when the pusher wire **102** is positioned through the sheath **106**.
(70) When the user is ready to advance the implant **104**, the locking element **108** may be removed proximally from the proximal region **129** of the sheath **106**. This may allow the inner diameter of the sheath **106** adjacent to the arms **120** to expand to the original configuration thus removing the friction lock and allowing the pusher wire **102** to move freely. Thus, as the sheath **106** is removed from the pusher wire **102** and/or the pusher wire **102** is distally advanced, the sheath **106** does not engage or hang up on any portion of the pusher wire **102**.
(71) It is contemplated that when the locking element **108** is provided with a longitudinally extending slot, such as the slot **310** illustrated in the locking element **300**, the locking element **108** may be free to drop off of the pusher wire **102** (e.g., via the slot) without having to be proximally retracted over an entirety of the length of the pusher wire **102** proximal to the sheath **106**. Similarly, during loading of the locking element **108** at the time of manufacture, the locking element **108** can be placed over the pusher wire **102** rather than threading the locking element **108** over the pusher wire **102**. As described above with respect to FIGS. **6** and **7**, if so provided a longitudinal slot may be sized such that the pusher wire **102** can fit through the slot, but not the sheath **106**. This may ensure the locking element **108** does not fall off the sheath **106** in the locked position (e.g., FIGS. **10** and **11**) but can be easily removed from the pusher wire **102** once the locking element is in the unlocked configuration.
(72) It is further contemplated that that locking element **108** may be formed from a material that is more rigid than or stiffer than the outer sheath **106**. For example, the material for the locking element **108** may be selected such that the locking element **108** does not deflect when the tapered portion of the lumen **135** of the locking element **108** is advanced over the sheath **106** but rather forces the sheath **106** to deflect inward. In some embodiments, the locking element **108** may be formed from a bright (or other) color easily noticeable by the user.
(73) FIG. **12** is a perspective view of another illustrative medical device system **400** in a partially unassembled configuration. FIG. **13** is a cross-sectional view of the illustrative medical device system **400** taken at line **13-13** of FIG. **12**. The medical device system **400** may include a pusher wire **402**, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath **406**, and a locking element **408**. For simplicity, the implant is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.
(74) Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **406** containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath **406** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **406** may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element **408** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **402** and implant within the sheath **406** until the user is ready to advance the implant out of the sheath **406**.
(75) The sheath **406** may be a tubular member including a proximal end **412**, a distal end **414**, and an intermediate region **415** positioned therebetween. Some suitable but non-limiting materials for the sheath **406**, for example, polymer materials, composite materials, etc., are described below. The

sheath **406** may define a lumen **410** extending from the proximal end **412** to the distal end **414**. The pusher wire **402** and implant may be slidably disposed within the lumen **410** of the sheath **406** such that the pusher wire **402** and the implant are radially inwards of the sheath **406**. The implant may be disposed proximate the distal end **414** of the sheath **406**. The pusher wire **402** may be axially slidable between an interlocked position and a released position. The pusher wire **402** may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **402** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **402**, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **402** may be releasably secured to the sheath **406** via the locking element **408** to limit axial and/or rotational movement of the pusher wire **402** within the sheath **406**.
(76) The sheath **406** may have a longitudinally extending slot **416** extending distally from the proximal end **412**. The slot **416** may terminate at a "C" cut **418** extending about a circumference of the sheath **416**. The cut **418** may be positioned in a plane that is generally perpendicular to the longitudinal axis of the sheath **406**. In some cases, the cut **418** may extend less than 360°, less than 270°, less than 180°, less than 135°, less than 90° about the circumference of the sheath **406**, as desired. It is contemplated that the radial length of the cut **418** may be varied to produce a desired locking effect. Similarly, the length and width of the slot **416** may be varied to produce a desired locking effect. The slot **416** and the cut **418** may allow the free edges or flaps **420***a*, **420***b* (collectively, **420**) of the sheath **406** to fold over or collapse when a radially inward force is applied to an outer surface thereof to create a frictional lock, as will be described in more detail here.
(77) It is contemplated that a proximal end region **422** of the sheath **406** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **422** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **406** may be formed from a same material as the proximal end region **422** while in other cases, other portions of the sheath **406** may be formed from a different material as the proximal end region **422**. For example, the sheath **406** may include a polyimide distal tip (although this is not required).
(78) Referring additionally to FIG. **14**, which illustrates a perspective view of the locking element **408**, the locking element **408** may be a tubular member having a proximal end **424**, a distal end **426**, and an intermediate region **428** positioned therebetween. The locking element **408** may have a generally constant or uniform outer diameter **430** over the intermediate region **428**. The outer diameter may increase from the intermediate region **428** in both the proximal and distal directions to create a flared proximal portion **432** and a flared distal portion **434**. The flared proximal portion **432** may increase in diameter towards the proximal end **424** to a second outer diameter **436** greater than the first outer diameter **430** adjacent to the intermediate region **428**. Similarly, the flared distal portion **434** may increase in diameter towards the distal end **426** to the second outer diameter **436** greater than the first outer diameter **430** adjacent to the intermediate region **428**.
(79) The locking element **408** may define a lumen **438** extending from the proximal end **424** to the distal end **426**. As will be discussed in more detail herein, a portion of the lumen **438** of the locking element **408** may be sized to slide freely over the sheath **406** while another portion of the lumen **438** may be sized to exert a radially inward compressing force on the sheath **406**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **406** may be radially inward of the locking element **408**.
(80) The locking element **408** may be formed in a variety of different manners. For example, the locking element **408** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **408** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.
(81) FIG. **15** is a cross-sectional view of the locking element **408** taken at line **15-15** of FIG. **14**. The locking element **408** may have a variable inner diameter. In some cases, the lumen **438** may

have a generally hourglass shape. For example, in some embodiments, the locking element **408** may have an inner diameter which has a first inner diameter **440** adjacent to the proximal and distal ends **424**, **426** thereof. The first inner diameter **440** may taper or slope towards a second generally constant inner diameter **442**. It is contemplated that the sloped inner diameter may generally correspond to the flared proximal and distal portions **432**, **434** while the generally constant inner diameter **442** may generally correspond to the intermediate region **428**. However, this is not required. However, in such an arrangement, the locking element **408** may have a generally uniform wall thickness from the proximal end **424** to the distal end **426** thereof. In other embodiments the inner diameter of the locking element **408** may be varied by varying a wall thickness of the locking element **408** along a length thereof. Further, in some cases, while not explicitly shown, the transition from the first inner diameter **440** to the second inner diameter **442** may be an abrupt or step-wise transition. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the flared proximal portion **432**, the flared distal portion **434** and/or the intermediate region **428** may be changed to create the desired locking effect (described in more detail herein).

(82) The first, larger inner diameter **440** may be sized such that the locking element **408** can slide freely over the sheath **406**. The first, smaller inner diameter **442** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **406**, as will be described in more detail herein.

(83) It is contemplated that the configuration of the locking element **408** may be adjusted to create the desired effect. For example, one or more of the inner diameters **440**, **442** may be made larger or smaller to accommodate different sizes of sheaths **406**. Further, the flared proximal portion **432**, the flared distal portion **434** and/or the intermediate region **428** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **408** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **408** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **406** and/or the locking element **408** may include visual indicia to guide the user in manipulation of the locking element **408**. In some cases, the overall length of the locking element **408** can be increased or decreased, as desired. It is further contemplated that the locking element **408** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(84) FIG. **16** is a perspective view of the illustrative medical device system **400** in a locked configuration and FIG. **17**A is a partial cross-sectional view of the illustrative medical device system **400**, taken at line **17**A-**17**A of FIG. **16**. While the medical device system **400** is described with respect to a particular locking element **408**, it should be understood that any of the locking elements described herein, including but not limited to the locking elements **108**, **200**, **300**, **508**, **600**, **708**, **800**, may be substituted for the locking element **408**. From the partially unassembled configuration illustrated in FIG. **12**, to position the locking element **408** such that it can be used to secure the pusher wire **402**, the locking element **408** is distally advanced over a proximal end region **422** of the sheath **406** from the proximal end **412**.

(85) To lock the pusher wire **402** relative to the sheath **406**, the locking element **408** is advanced further in the distal direction until the intermediate region **428** of the locking element **408** is disposed over the flaps **420** of the sheath **406**. The flared distal portion **432** of locking element **408** may slide freely over the proximal end region **422** of the sheath. When the intermediate region **428** having the reduced diameter **442** of the locking element **408** reaches the outer sheath **406**, the flaps **420** of the sheath **406** may fold over and be depressed onto an outer surface of the pusher wire **402**. This is more clearly illustrated in FIG. **17**B, which is a partial cross-section view of the illustrative medical device system **400**, taken at line **17**B-**17**B of FIG. **17**A. For example, the locking element **408** may freely slide over the outer sheath **406** until the inner diameter of the locking element **408** is approximately equal to or less than an outer diameter of the sheath **406**. Once the inner diameter

of the locking element **408** begins to frictionally engage the outer surface of the sheath **406** more force may be required to continue distal movement of the locking element **408**. As the locking element **408** is distally advanced further onto the sheath **406**, the intermediate region **428** acts to deflect the flaps **420** of the sheath **406** inward and onto the wire **402**. In some cases, the tapered lumen of the flared distal portion **434** may gradually deflect the flaps **420** inward. When advanced far enough, the locking element **408** wedges in place, creating the desired locking effect between the sheath **406** and the pusher wire **402**. For example, the reduced diameter portion (e.g., having the second smaller inner diameter **442**) of the lumen **438** of the locking element **408** may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the flaps **420** of the sheath **406** such that the outer diameter of the flaps **420** is decreased and the inner diameter of the arms is biased radially inward, as shown in FIG. **17**B. As the inner diameter of the flaps **420** is reduced, the inner surface of the flaps **420** contacts and frictionally engages an outer surface of the pusher wire **402**.

(86) In the locked configuration illustrated in FIGS. **16**, **17**A, and **17**B, the frictional engagement between the inner surface of the sheath **406** and the outer surface of the pusher wire **402** may preclude or inhibit axial (e.g., proximal and distal) and/or rotational movement of the pusher wire **402** within the lumen **410** of the sheath **406**. In some embodiments, the flaps **420** are configured to promote inward deflection thereof. For example, the cut pattern of the slot **416** and/or slit **418** may be applied to promote collapse of the flaps **420** and to create a localized friction point between an inner surface of the sheath **406** (e.g., the inner surface of the flaps **420**) and the pusher wire **402**.

(87) It is contemplated that the gradual decrease in the inner diameter of the locking element **408** may facilitate positioning of the locking element **408** over the sheath **406**. For example, the inner diameter **440** of the locking element **408** at the distal portion **434** (and/or portions of the proximal portion **432** may be large enough to freely pass over the outer diameter of the sheath **406**. As the inner diameter of the locking element **408** tapers (e.g., in the proximal direction towards the proximal end **424**) to a dimension smaller than the outer diameter of the sheath **406**, and is distally advanced far enough over the sheath **406**, it promotes collapse of the flaps **420**. This creates a friction lock between the sheath **406** and pusher wire **402** when the pusher wire **402** is positioned through the sheath **406**.

(88) When the user is ready to advance the implant, the locking element **408** may be removed proximally from the proximal region **422** of the sheath **406**. This may allow the inner diameter of the sheath **406** adjacent to the flaps **420** to expand to the original configuration thus removing the friction lock and allowing the pusher wire **402** to move freely. Thus, as the sheath **406** is removed from the pusher wire **402**, the sheath **406** does not engage or hang up on any portion of the pusher wire **402**.

(89) It is contemplated that when the locking element **408** is provided with a longitudinally extending slot, such as the slot **310** illustrated in the locking element **300**, the locking element **408** may be free to drop off of the pusher wire **402** (e.g., via the slot) without having to be proximally retracted over an entirety of the length of the pusher wire **402** proximal to the sheath **406**. Similarly, during loading of the locking element **408** at the time of manufacture, the locking element **408** can be placed over the pusher wire **402** rather than threading the locking element **408** over the pusher wire **402**. As described above with respect to FIGS. **6** and **7**, if so provided, a longitudinal slot may be sized such that the pusher wire **402** can fit through the slot, but not the sheath **406**. This may ensure the locking element **408** does not fall off the sheath **406** in the locked position (e.g., FIGS. **16**, **17**A, and **17**B) but can be easily removed from the pusher wire **402** once the locking element is in the unlocked configuration.

(90) It is further contemplated that that locking element **408** may be formed from a material that is more rigid than or stiffer than the outer sheath **406**. For example, the material for the locking element **408** may be selected such that the locking element **408** does not deflect when the tapered portion of the lumen **438** of the locking element **408** is advanced over the sheath **406** but rather

forces the sheath **406** to deflect inward. In some embodiments, the locking element **408** may be formed from a bright (or other) color easily noticeable by the user.

(91) FIG. **18** is a perspective view of an example medical device system **500** in an unlocked assembled configuration. FIG. **19** is a partial cross-sectional view of the medical device system **500** taken at line **19-19** of FIG. **18**. The medical device system **500** may include a pusher wire **502**, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath **506**, and a locking element **508**. For simplicity, the implant is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.

(92) Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **506** containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath **506** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **506** may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element **508** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **502** and implant within the sheath **506** until the user is ready to advance the implant out of the sheath **506**.

(93) The sheath **506** may be a tubular member including a proximal end **512** (e.g., FIG. **19**), a distal end **514**, and an intermediate region **516** positioned therebetween. Some suitable but non-limiting materials for the sheath **506**, for example, polymer materials, composite materials, etc., are described below. The sheath **506** may define a lumen **510** extending from the proximal end **512** to the distal end **514**. The pusher wire **502** and implant may be slidably disposed within the lumen **510** of the sheath **506** such that the pusher wire **502** and the implant are radially inwards of the sheath **506**. The implant may be disposed proximate the distal end **514** of the sheath **506**. The pusher wire **502** may be axially slidable between an interlocked position and a released position. The pusher wire **502** may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **502** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **502**, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **502** may be releasably secured to the sheath **506** via the locking element **508** to limit axial and/or rotationally movement of the pusher wire **502** within the sheath **506**.

(94) It is contemplated that a proximal end region **520** of the sheath **506** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **520** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **506** may be formed from a same material as the proximal end region **520** while in other cases, other portions of the sheath **506** may be formed from a different material as the proximal end region **520**. For example, the sheath **506** may include a polyimide distal tip (although this is not required).

(95) Referring additionally to FIG. **19**, the sheath **506** may include an angled protrusion **518** adjacent the proximal end **512** and extending radially outward from an outer surface of the sheath **506**. The angled protrusion **518** may cooperate with a corresponding protrusion **528** on the locking element **508** to allow the locking element **508** to be distally advanced over the sheath **506** while providing a mechanical stop that limits actuation of the locking element **508** in the proximal direction. In some instances, the angled protrusion **518** may be a single component extending about an entire circumference (e.g., 360°) of the sheath **506**. In other cases, the angled protrusion **518** may be formed from a plurality of individual protrusions which are spaced (uniformly or eccentrically) about the circumference of the sheath **506**. The angled protrusion **518** may have an

outer diameter which decreases in the proximal direction (e.g., towards the proximal end **512**). The distal end of the angled protrusion **518** may be generally planar and have an outer diameter that is greater than an inner diameter of the proximal end of the protrusion **528** of the locking element **508**, as will be described in more detail herein. The distal end of the angled protrusion **518** may extend generally orthogonal to a longitudinal axis of the sheath **506** while the angled surface extends at an angle that is non-orthogonal and non-parallel to the longitudinal axis of the sheath **506**.

(96) The locking element **508** may be a tubular member having a proximal end **522**, a distal end **524**, and an intermediate region **526** positioned therebetween. The locking element **508** may have a tapered or sloped outer diameter over a proximal region **536** that increases from the proximal end **522** towards and into the intermediate region **526**. The locking element **508** may include a distal region **538** having a generally uniform or constant outer diameter. The locking element **508** may define a lumen **530** extending from the proximal end **522** to the distal end **524**. As will be discussed in more detail herein, a portion of the lumen **530** of the locking element **508** may be sized to slide freely over the sheath **506** while another portion of the lumen **530** may be sized to exert a radially inward compressing force on the sheath **506**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **506** may be radially inward of the locking element **508**.

(97) The locking element **508** may be formed in a variety of different manners. For example, the locking element **508** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **508** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

(98) The locking element **508** may have a variable inner diameter. For example, in some embodiments, the locking element **508** may have an inner diameter which increases in a sloped manner over the proximal region **536** in the distal direction from a first, smaller, inner diameter adjacent the proximal end **522** to a second, larger, inner diameter. The second inner diameter **138** may be approximately constant or uniform over a distal region **538** of the locking element **508**. In some cases, while not explicitly shown, the transition from the first inner diameter to the second inner diameter **138** may be an abrupt or step-wise transition. The proximal region **536** and/or the distal region **538** may vary, as desired. Further, the slope of the proximal region **536** may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered proximal region **536** and/or the distal region **538** may be changed to create the desired locking effect (described in more detail herein).

(99) In some embodiments, a wall thickness of the locking element **508** may remain generally uniform over the proximal and/or distal regions **536**, **538** of the locking element **508**. In such an instance, the inner diameter and the outer diameter of the locking element **508** may vary (e.g., may be sloped in a similar manner) along the proximal region **536**. For example, the outer diameter of the locking element **508** may increase from a first outer diameter at the proximal end **522** to a second outer diameter (greater than the first outer diameter) over the proximal region **536**, as the inner diameter increases. In some embodiments, the wall thickness may vary over the proximal and/or distal regions **536**, **538** of the locking element **508**. The second, larger, inner diameter may be sized such that the locking element **508** can slide freely over the sheath **506**. The first, smaller, inner diameter may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **506**, as will be described in more detail herein.

(100) The locking element **508** may include an angled protrusion **528** adjacent the distal end **524** and extending radially inward from an inner surface of the locking element **508**. The angled protrusion **528** may cooperate with the protrusion **518** on the sheath **506** to allow the locking element **508** to be distally advanced over the sheath **506** while providing a mechanical stop that limits actuation of the locking element **508** in the proximal direction. In some instances, the angled protrusion **528** may be a single component extending about an entire inner surface of the lumen **530**

(e.g., 360°) of the locking element **508**. In other cases, the angled protrusion **528** may be formed from a plurality of individual protrusions which are spaced (uniformly or eccentrically) about the inner surface of the lumen **530**. The angled protrusion **528** may have an inner diameter which decreases in the distal direction (e.g., towards the distal end **524**). The proximal end of the angled protrusion **528** may be generally planar and have an inner diameter that is less than an outer diameter of the distal end of the protrusion **518** of the sheath **506**. The proximal end of the angled protrusion **528** may extend generally orthogonal to a longitudinal axis of the locking element **508** while the angled surface extends at an angle that is non-orthogonal and non-parallel to the longitudinal axis of the locking element **508**.

(101) The sloped surfaces of the angled protrusions **518**, **528** may be structured such that the angled protrusion **518** on the sheath **506** essentially functions as a ramp for the angled protrusion **528** on the locking element **508**. This may allow the locking element **508** to be loaded onto the proximal end region **520** of the sheath **506** by advancing the distal end **512** of the sheath **506** through the distal end **524** of the locking element. Once loaded onto the sheath **506**, proximal retraction of the locking element **508** will be limited. For example, the generally planar proximal end of the angled protrusion **528** on the locking element **508** will abut the generally planar distal end of the angled protrusion **518** on the sheath **506** creating a mechanical stop thus limiting the axial movement of the locking element **508**. This may allow the locking element **508** to be moved between the locked (see, for example, FIG. **20**) and unlocked configuration over a shorter distance and without fully removing it from the pusher wire **502**.

(102) It is contemplated that the configuration of the locking element **508** may be adjusted to create the desired effect. For example, one or more of the inner diameters may be made larger or smaller to accommodate different sizes of sheaths **506**. Further, the proximal and/or distal regions **536**, **538** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **508** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **508** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **506** and/or the locking element **508** may include visual indicia to guide the user in manipulation of the locking element **508**. In some cases, the overall length of the locking element **508** can be increased or decreased, as desired. It is further contemplated that the locking element **508** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(103) While the medical device system **500** is described with respect to a particular locking element **508**, it should be understood that any of the locking elements described herein may be substituted for the locking element **508**. To position the locking element **508** such that it can be used to secure the pusher wire **502**, the locking element **508** is distally advanced over the proximal end region **520** of the sheath **506** from the proximal end **512**. It should be understood that the varying lengths of the locking element **508** may be positioned over the sheath **506** in an unlocked configuration. However, an entirety of the locking element **508** may not be disposed over sheath **506** in the unlocked configuration. For example, the proximal region **536** of the locking element **508** having the reduced inner diameter may be only partially disposed over the sheath **506** or not over the sheath **506** at all. In this configuration, the pusher wire **502** is free to slide axially (e.g., proximally and distally) and/or rotate within the lumen **510** of the sheath **506**. As described above, the inner diameter of the sheath **506** may remain constant, or substantially constant from the proximal end **512** to the distal end **514**.

(104) To lock the pusher wire **502** relative to the sheath **506**, the locking element **508** is advanced further in the distal direction until the proximal region **536** of the locking element **508** is disposed over the proximal end region **520** of the sheath **506**. FIG. **20** is a partial cross-sectional view of the medical device system **500** in a locked configuration. As described above, the distal portion **538** and a portion of the proximal portion **536** of locking element **508** may slide freely over the

proximal end region **520** of the sheath. When proximal portion **536** having the reduced diameter of the locking element **508** reaches the outer sheath **506**, the sheath **506** may be depressed onto an outer surface of the pusher wire **502**. For example, the locking element **508** may freely slide over the outer sheath **506** until the inner diameter of the locking element **508** is approximately equal to or less than an outer diameter of the sheath **506**. Once the inner diameter of the locking element **508** begins to frictionally engage the outer surface of the sheath **506** more force may be required to continue distal movement of the locking element **508**. As the locking element **508** is distally advanced further onto the sheath **506**, the tapered region (e.g., the proximal region **536**) acts to deflect the sheath **506** inward and onto the wire **502**. When advanced far enough, the locking element **508** wedges in place, creating the desired locking effect between the sheath **506** and the pusher wire **502**. For example, the tapered portion of the lumen **530** of the locking element **508** may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the sheath **506** such that the outer diameter of sheath **506** is decreased and the inner diameter of the sheath **506** is biased radially inward, as shown in FIG. **20**. As the inner diameter of the sheath **506** is reduced, the inner surface of the sheath **506** contacts and frictionally engages an outer surface of the pusher wire **502**.

(105) In the locked configuration illustrated in FIG. **20**, the frictional engagement between the inner surface of the sheath **506** and the outer surface of the pusher wire **502** may preclude or inhibit axial (e.g., proximally and distally) and/or rotational movement of the pusher wire **502** within the lumen **510** of the sheath **506**. In some embodiments, the sheath **506** may configured to promote inward deflection thereof. For example, the sheath **506** may include cuts, slots, have a reduced thickness, be more flexible, etc. to promote collapse of the sheath **506** and to create a localized friction point between an inner surface of the sheath **506** and the pusher wire **502**.

(106) It is contemplated that the gradual decrease in the inner diameter of the locking element **508** may facilitate positioning of the locking element **508** over the sheath **506**. For example, the inner diameter of the locking element **508** at the distal portion **538** (and/or portions of the proximal portion **536** may be large enough to freely pass over the outer diameter of the sheath **506**. As the inner diameter of the locking element tapers (e.g., in the proximal direction towards the proximal end **522**) to a dimension smaller than the outer diameter of the sheath **506**, and is distally advanced far enough over the sheath **506**, it promotes collapse of the sheath **506**. This creates a friction lock between the sheath **506** and pusher wire **502** when the pusher wire **502** is positioned through the sheath **506**.

(107) When the user is ready to advance the implant, the locking element **508** may be proximally displaced (but not entirely removed) from the proximal region **520** of the sheath **506**. This may allow the inner diameter of the sheath **506** to expand to the original configuration thus removing the friction lock and allowing the pusher wire **502** to move freely. Thus, as the sheath **506** is removed from the pusher wire **502**, the sheath **506** does not engage or hang up on any portion of the pusher wire **502**.

(108) It is contemplated that that locking element **508** may be formed from a material that is more rigid than or stiffer than the outer sheath **506**. For example, the material for the locking element **508** may be selected such that the locking element **508** does not deflect when the tapered portion of the lumen **530** of the locking element **508** is advanced over the sheath **506** but rather forces the sheath **506** to deflect inward. In some embodiments, the locking element **508** may be formed from a bright (or other) color easily noticeable by the user.

(109) FIG. **21** illustrates a perspective view of another illustrative locking element **600**. The locking element **600** may be similar in form and function to the illustrative locking mechanism **108** described herein and may be used with a medical device system, such as the medical device system **100** described herein. The locking element **600** may be a tubular member having a proximal end **602**, a distal end **604**, and an intermediate region **605** positioned therebetween. The locking element **600** may define a lumen **612** extending from the proximal end **602** to the distal end **604**. A

portion of the lumen **612** of the locking element **600** may be sized to slide freely over a sheath, such as the sheath **106** described herein, while another portion of the lumen **612** may be sized to exert a radially inward or compressive force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **600**.

(110) The locking element **600** may have a tapered or sloped outer diameter over a proximal region **628** extending from the proximal end **602** towards and into the intermediate region **605**. Similarly, the locking element **600** may have a tapered or sloped outer diameter over a distal region **630** extending from the distal end **604** towards and into the intermediate region **605**. However, in some cases, one or both the proximal region **628** and the distal region **630** may have a generally constant outer diameter or the diameter may increase towards the intermediate region **605** as desired. The intermediate region **605** may include plurality of raised regions or circumferential ribs **632***a*, **632***b*, **632***c* (collectively, **632**). The circumferential ribs **632** may have an outer diameter **634** that is greater than an outer diameter **624** of the intermediate region **605** between the ribs **632**. The circumferential ribs **632** may extend about an entire perimeter of the locking element **600** or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs **632** may increase the tactility of the locking element **600** and/or make the locking element **600** easier to grip or handle. The number of circumferential ribs **632** (e.g., fewer than three or greater than three), the size of the circumferential ribs **632** (e.g., increasing or decreasing the diameter **634**), the geometry of the circumferential ribs **632** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **632** can be varied to enhance or diminish the tactility or gripability of the locking element **600**, as desired.

(111) The locking element **600** may be formed in a variety of different manners. For example, the locking element **600** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **600** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

(112) FIG. **22** is a cross-sectional view of the locking element **600** taken at line **22-22** of FIG. **21**. The locking element **600** may have a variable outer diameter and a variable inner diameter. For example, in some embodiments, the locking element **600** may have an inner diameter which increases in a sloped manner in the distal direction from a first inner diameter **618** adjacent the proximal end **602** to a second inner diameter **620** adjacent the distal end **604**. Further, the slope of the inner surface may vary, as desired. The second, larger inner diameter **620** may be sized such that the distal region **630** of the locking element **600** can slide freely over the sheath **106**. The first, smaller inner diameter **618** may be sized such that the proximal region **628** is configured to apply a compressive, or radially inward force on the outer surface of the sheath **106**.

(113) In some embodiments, a wall thickness **636** of the locking element **600** may vary over a length of the locking element **600**. In some cases, the wall thickness **636** is not directly correlated to the inner diameter of the locking element **600**. In other cases, the wall thickness **636** may reduce in size as the inner diameter increases. These are just some examples. The outer diameter of the locking element **600** may decrease from a first outer diameter **624** at the proximal end **602** to a second outer diameter **626** (less than the first outer diameter **624**) adjacent to the intermediate region **605**. The outer diameter of the locking element **600** may decrease from a third outer diameter **636** at the proximal end **602** to a fourth outer diameter **632** (less than the third outer diameter **636**) adjacent to the intermediate region **605**. In some embodiments the first outer diameter **622** and the third outer diameter **636** may be approximately equal, although this is not required. Similarly, the second outer diameter **626** and the fourth outer diameter **632** may be approximately equal, although this is not required.

(114) It is contemplated that the configuration of the locking element **600** may be adjusted to create the desired effect. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered inner diameter may be changed to create the desired locking effect. For example, one or

more of the inner diameters **618**, **620** may be made larger or smaller to accommodate different sizes of sheaths **106**. In another example, the outer diameter of the locking element **600** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **600** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **600** may include visual indicia to guide the user in manipulation of the locking element **600**. In some cases, the overall length of the locking element **600** can be increased or decreased, as desired. It is further contemplated that the locking element **600** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(115) FIG. **23** is a perspective view of another illustrative medical device system **700** in a partially unassembled configuration. The medical device system **700** may include a pusher wire **702**, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath **706**, and a locking element **708**. For simplicity, the implant is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc. While the medical device system **400** is described with respect to a particular locking element **708**, it should be understood that any of the locking elements described herein, including but not limited to the locking elements **108**, **200**, **300**, **408**, **508**, **600**, **800**, may be substituted for the locking element **408**.

(116) Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **706** containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath **706** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **706** may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element **708** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **702** and implant within the sheath **706** until the user is ready to advance the implant out of the sheath **706**.

(117) The sheath **706** may be a tubular member including a proximal end **712**, a distal end **714**, and an intermediate region **716** positioned therebetween. Some suitable but non-limiting materials for the sheath **706**, for example, polymer materials, composite materials, etc., are described below. The sheath **706** may define a lumen **710** extending from the proximal end **712** to the distal end **714**. The pusher wire **702** and implant may be slidably disposed within the lumen **710** of the sheath **706** such that the pusher wire **702** and the implant are radially inwards of the sheath **706**. The implant may be disposed proximate the distal end **714** of the sheath **706**. The pusher wire **702** may be axially slidable between an interlocked position and a released position. The pusher wire **702** may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **702** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **702**, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **702** may be releasably secured to the sheath **706** via the locking element **708** to limit axial and/or rotational movement of the pusher wire **702** within the sheath **706**.

(118) The sheath **706** may have a first slot **718***a* and a second slot **718***b* (collectively, **718**) extending distally from the proximal end **712**. The first and second slots **718***a*, **718***b* may be positioned across from another or spaced approximately 180° about the circumference of the sheath **706**. While the sheath **706** is described as having two slots, the sheath **706** may include fewer than

two or more than two slots, as desired. The slots **718** may extend less than an entire length of the sheath **706**. The slots **718** may remove material from the sheath **706** to create flexible arms or members **720***a*, **720***b* (collectively, **720**). While the sheath **706** is described as having two arms **720**, it should be understood the number of flexible arms may vary with the number of slots **718** and there can be fewer than two or more than two flexible arms, as desired. Further, the slots **718** may be uniformly or eccentrically distributed about a circumference of the sheath **706**. The length and/or size of the slots **718** (and/or the arms **720**) may vary to produce different degrees of wedging (between the locking element **708** and the pusher wire **702**) and locking capabilities.
(119) The sheath **706** may further include a raised projection **730** extending radially from an outer surface thereof. The raised projection **730** may be positioned adjacent to the proximal end region **729** of the sheath **706**. In some cases, the raised projection **730** may extend distally from the proximal end **712** of the sheath **706**. However, this is not required. The raised projection **730** may be a rectangular prism, such as, but not limited to a ridge or other three-dimensional structure configured to engage a mating groove, notch, or recess within the locking element **708**, as will be described in more detail herein.
(120) It is contemplated that a proximal end region **729** of the sheath **706** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **729** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **706** may be formed from a same material as the proximal end region **729** while in other cases, other portions of the sheath **706** may be formed from a different material as the proximal end region **729**. For example, the sheath **706** may include a polyimide distal tip (although this is not required).
(121) Referring additionally to FIG. **24**, which illustrates a perspective view of the locking element **708**, the locking element **708** may be a tubular member having a proximal end **722**, a distal end **724**, and an intermediate region **726** positioned therebetween. The locking element **708** may have a generally hourglass shaped outer profile. For example, the outer diameter may generally decrease from the proximal end **722** towards the intermediate region **726** and increase from the intermediate region **726** to the distal end **724**. However, this is not required. The outer profile of the locking element **708** may take any shape desired.
(122) The locking element **708** may define a lumen **728** extending from the proximal end **722** to the distal end **724**. As will be discussed in more detail herein, a portion of the lumen **728** of the locking element **708** may be sized to slide freely over the sheath **706** while another portion of the lumen **728** may be sized to exert a radially inward compressing force on the sheath **706**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **706** may be radially inward of the locking element **708**.
(123) The intermediate region **726** may include plurality of raised regions or circumferential ribs **732***a*, **732***b*, **732***c* (collectively, **732**). The circumferential ribs **732** may have an outer diameter **734** that is greater than an outer diameter **736** of the intermediate region **726** between the ribs **732**. The circumferential ribs **732** may extend about an entire perimeter of the locking element **708** or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs **732** may increase the tactility of the locking element **708** and/or make the locking element **708** easier to grip or handle. The number of circumferential ribs **732** (e.g., fewer than three or greater than three), the size of the circumferential ribs **732** (e.g., increasing or decreasing the diameter **734**), the geometry of the circumferential ribs **732** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **732** can be varied to enhance or diminish the tactility or gripability of the locking element **708**, as desired.
(124) The locking element **708** may further include one or more features to provide a visual cue as to how the locking element **708** should be assembled with the sheath **706**. For example, the feature may be configured to provide an indication to the user how to align the locking element **708** with the sheath **706** such that the locking element **708** may be slid over the sheath **706**. In some cases, the feature may be incorporated into the structure of the locking element **708**. For example, in the

embodiment illustrated in FIG. **24**, the locking element **708** includes an axially extending (e.g., parallel to a longitudinal axis of the locking element **708**) raised portion **738**. The raised portion **738** is radially aligned with an internal groove **740** (see, for example, FIGS. **25** and **26**) formed in an inner surface of the locking element **708**. As will be described in more detail herein, the groove **740** may be configured to align with the raised projection **730** of the sheath **706** to allow for selective assembly of the locking element **708** with the sheath **706**. The raised portion **738** is just one example of a feature that may be used to provide visual cues to the operator. Other features may include visual markings on an outer surface of the locking element **708** (e.g., arrows, words, etc.) or other structural features, as desired.

(125) The locking element **708** may be formed in a variety of different manners. For example, the locking element **708** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **708** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc. It is further contemplated that that locking element **708** may be formed from a material that is more rigid than or stiffer than the outer sheath **706**. For example, the material for the locking element **708** may be selected such that the locking element **708** does not deflect when the tapered portion of the lumen **728** of the locking element **708** is advanced over the sheath **706** but rather forces the sheath **706** to deflect inward. In some embodiments, the locking element **708** may be formed from a bright (or other) color easily noticeable by the user.

(126) FIG. **25** is a cross-sectional view of the locking element **708** taken at line **25-25** of FIG. **24**. FIG. **26** is a distal end view of the locking element **708**. The locking element **708** may have a variable inner cross-sectional dimension. In some cases, the lumen **728** may have a generally tapered configuration. For example, in some embodiments, the lumen **728** of the locking element **708** may have a first cross-sectional dimension **742** adjacent to the proximal end **722** thereof which increases in the distal direction to a second cross-sectional dimension **744** adjacent a distal end **724** thereof which is greater than the first cross-sectional dimension **742**. The cross-sectional dimension may gradually transition in a tapered or sloped manner. In other embodiments, the cross-sectional dimension may vary in an abrupt or step-wise manner. These are just some examples. In some cases, a wall thickness of the locking element **708** may vary over at least a portion of the locking element. However, this is not required. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the locking element **708** may be changed to create the desired locking effect (described in more detail herein). The distal inner cross-sectional dimension **744** may be sized such that the locking element **708** can slide freely over the sheath **706**. The proximal, smaller inner cross-sectional dimension **742** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **706**.

(127) It is contemplated that the configuration of the locking element **708** may be adjusted to create the desired effect. For example, one or more of the inner cross-sectional dimensions **742**, **744** may be made larger or smaller to accommodate different sizes of sheaths **706**. In another example, the outer diameter of the locking element **708** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **708** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **706** and/or the locking element **708** may include visual indicia to guide the user in manipulation of the locking element **708**. In some cases, the overall length of the locking element **708** can be increased or decreased, as desired. It is further contemplated that the locking element **708** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

(128) As described above, the locking element **708** may further include an axially extending (e.g., parallel to a longitudinal axis of the locking element) notch or groove **740**. As shown in FIG. **25**, the groove **740** may have a first depth **746** adjacent to the distal end **724** and a second depth **748** adjacent to the proximal end **722**. The second depth **748** is less than the first depth **746** such that a distal portion of the locking element **708** can freely slide over the sheath **706** when the groove **740**

is aligned with the raised projection **730** of the sheath **706**. As the locking element **708** is distally advanced, the lumen **728** exerts a radially inward or compressive force on the sheath **706**.

(129) Together, the radial projection **730** of the sheath **706** and the groove **740** of the locking element **708** may create an orientation dependent assembly. For example, the locking element **708** is oriented such that the groove **740** is aligned with the radial projection **730** of the sheath to allow the locking element **708** to distally advance over the sheath **706**. Misalignment of the groove **740** and radial projection **730** may lead to an interference between the locking element **708** and the sheath **706** that would prevent passage of the sheath **706** into the locking element **708** and subsequent locking of the pusher wire **702**. This may prevent undesired re-engagement of the locking element **708** and sheath **706** during use of the system **700** in the event the locking element **708** is not fully removed from the pusher wire **702**. In the absence of orientation dependent features, if the locking element **708** is left hanging on the pusher wire **702**, it may unintendedly re-engage with the sheath **706** and the pusher wire **702** during advancement of the pusher wire **702** through the sheath **706**. It is contemplated that other orientation dependent features may be used on the sheath **706** and/or the locking element **708**, as desired.

(130) FIG. **27** illustrates a perspective view of another illustrative locking element **800** which may include features configured to orientate the locking element **800** relative to a sheath, such as, but not limited to, the sheaths **106**, **406**, **506**, **706**, **706** described herein. The locking element **800** may be a tubular member having a proximal end **802**, a distal end **804**, and an intermediate region **806** positioned therebetween. The locking element **800** may have a generally hourglass shaped outer profile. For example, the outer diameter may generally decrease from the proximal end **802** towards the intermediate region **806** and increase from the intermediate region **806** to the distal end **804**. However, this is not required. The outer profile of the locking element **800** may take any shape desired.

(131) The locking element **800** may define a lumen **808** extending from the proximal end **802** to the distal end **804**. A portion of the lumen **808** of the locking element **800** may be sized to slide freely over a sheath while another portion of the lumen **808** may be sized to exert a radially inward compressing force on the sheath. Thus, in some configurations, at least a portion of the sheath **706** may be radially inward of the locking element **800**.

(132) The intermediate region **806** may include plurality of raised regions or circumferential ribs **810***a*, **810***b*, **810***c* (collectively, **810**). The circumferential ribs **810** may have an outer diameter **812** (see, for example, FIG. **28**) that is greater than an outer diameter **814** (see, for example, FIG. **28**) of the intermediate region **806** between the ribs **810**. The circumferential ribs **810** may extend about an entire perimeter of the locking element **800** or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs **810** may increase the tactility of the locking element **800** and/or make the locking element **800** easier to grip or handle. The number of circumferential ribs **810** (e.g., fewer than three or greater than three), the size of the circumferential ribs **810** (e.g., increasing or decreasing the diameter **812**), the geometry of the circumferential ribs **810** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **810** can be varied to enhance or diminish the tactility or gripability of the locking element **800**, as desired.

(133) The locking element **800** may be formed in a variety of different manners. For example, the locking element **800** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **800** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc. It is further contemplated that that locking element **800** may be formed from a material that is more rigid than or stiffer than the outer sheath **706**. For example, the material for the locking element **800** may be selected such that the locking element **800** does not deflect when the tapered portion of the lumen **808** of the locking element **800** is advanced over the sheath **706** but rather forces the sheath **706** to deflect inward. In some embodiments, the locking element **800** may be formed from a bright (or other) color easily noticeable by the user.

(134) FIG. **28** is a distal end perspective cross-sectional view of the locking element **800** taken at line **28-28** of FIG. **27**. FIG. **29** is a distal end view of the locking element **800**. The locking element **800** may have a variable inner cross-sectional dimension. In some cases, the lumen **808** may have a generally tapered configuration. For example, in some embodiments, the lumen **808** of the locking element **800** may have a first cross-sectional dimension **816** adjacent to the proximal end **802** thereof which increases in the distal direction to a second cross-sectional dimension **818** adjacent a distal end **804** thereof which is greater than the first cross-sectional dimension **816**. The cross-sectional dimension may gradually transition in a tapered or sloped manner. In other embodiments, the cross-sectional dimension may vary in an abrupt or step-wise manner. These are just some examples. In some cases, a wall thickness of the locking element **800** may vary over at least a portion of the locking element. However, this is not required. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the locking element **800** may be changed to create the desired locking effect (described in more detail herein). The distal inner cross-sectional dimension **818** may be sized such that the locking element **800** can slide freely over the sheath. The proximal, smaller inner cross-sectional dimension **816** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath.
(135) It is contemplated that the configuration of the locking element **800** may be adjusted to create the desired effect. For example, one or more of the inner cross-sectional dimensions **816**, **818** may be made larger or smaller to accommodate different sizes of sheaths. In another example, the outer diameter of the locking element **800** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **800** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath and/or the locking element **800** may include visual indicia to guide the user in manipulation of the locking element **800**. In some cases, the overall length of the locking element **800** can be increased or decreased, as desired. It is further contemplated that the locking element **800** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.
(136) The locking element **800** may further include an axially extending (e.g., parallel to a longitudinal axis of the locking element) protrusion **820** which extends radially inward from an inner surface of the locking element **80**. The protrusion **820** may be a rectangular prism, such as, but not limited to a ridge or other three-dimensional structure configured to engage a mating groove, notch, or recess within the sheath. The protrusion **820** may be sized and shaped such that a distal portion of the locking element **800** can freely slide over the sheath when the protrusion **820** is aligned with a mating slot or recess of the sheath. As the locking element **800** is distally advanced, the lumen **808** narrows so as to exert a radially inward or compressive force on the sheath.
(137) Together, the slot of the sheath and the protrusion **820** of the locking element **800** may create an orientation dependent assembly. For example, the locking element **800** is oriented such that the protrusion **820** is aligned with the slot of the sheath to allow the locking element **800** to distally advance over the sheath. Misalignment of the protrusion **820** and groove may lead to an interference between the locking element **800** and the sheath that would prevent passage of the sheath into the locking element **800** and subsequent locking of the pusher wire. This may prevent undesired re-engagement of the locking element **800** and sheath during use of the system in the event the locking element **800** is not fully removed from the pusher wire. In the absence of orientation dependent features, if the locking element **800** is left hanging on the pusher wire, it may unintendedly re-engage with the sheath and the pusher wire during advancement of the pusher wire through the sheath. It is contemplated that other orientation dependent features may be used on the sheath and/or the locking element **800**, as desired.
(138) While not explicitly shown, the locking element **800** may further include one or more features to provide a visual cue as to how the locking element **800** should be assembled with the sheath. For example, the features may include, but are not limited to, visual markings on an outer

surface of the locking element **800** (e.g., arrows, words, etc.) or structural features, as desired.

(139) In some embodiments, the medical device systems **100**, **400**, **500**, **700** the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable metals and metal alloys include stainless steel, such as 444V, 444L, and 314LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; platinum; palladium; gold; combinations thereof; and the like; or any other suitable material.

(140) As alluded to herein, within the family of commercially available nickel-titanium or nitinol alloys, is a category designated “linear elastic” or “non-super-elastic” which, although may be similar in chemistry to conventional shape memory and super elastic varieties, may exhibit distinct and useful mechanical properties. Linear elastic and/or non-super-elastic nitinol may be distinguished from super elastic nitinol in that the linear elastic and/or non-super-elastic nitinol does not display a substantial “superelastic plateau” or “flag region” in its stress/strain curve like super elastic nitinol does. Instead, in the linear elastic and/or non-super-elastic nitinol, as recoverable strain increases, the stress continues to increase in a substantially linear, or a somewhat, but not necessarily entirely linear relationship until plastic deformation begins or at least in a relationship that is more linear than the super elastic plateau and/or flag region that may be seen with super elastic nitinol. Thus, for the purposes of this disclosure linear elastic and/or non-super-elastic nitinol may also be termed “substantially” linear elastic and/or non-super-elastic nitinol.

(141) In some cases, linear elastic and/or non-super-elastic nitinol may also be distinguishable from super elastic nitinol in that linear elastic and/or non-super-elastic nitinol may accept up to about 2-5% strain while remaining substantially elastic (e.g., before plastically deforming) whereas super elastic nitinol may accept up to about 8% strain before plastically deforming. Both of these materials can be distinguished from other linear elastic materials such as stainless steel (that can also be distinguished based on its composition), which may accept only about 0.2 to 0.44 percent strain before plastically deforming.

(142) In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy is an alloy that does not show any martensite/austenite phase changes that are detectable by differential scanning calorimetry (DSC) and dynamic metal thermal analysis (DMTA) analysis over a large temperature range. For example, in some embodiments, there may be no martensite/austenite phase changes detectable by DSC and DMTA analysis in the range of about −60 degrees Celsius (° C.) to about 120° C. in the linear elastic and/or non-super-elastic nickel-titanium alloy. The mechanical bending properties of such material may therefore be generally inert to the effect of temperature over this very broad range of temperature. In some embodiments, the mechanical bending properties of the linear elastic and/or non-super-elastic nickel-titanium alloy at ambient or room temperature are substantially the same as the mechanical properties at body temperature, for example, in that they do not display a super-elastic plateau and/or flag region. For example, across

a broad temperature range, the linear elastic and/or non-super-elastic nickel-titanium alloy maintains its linear elastic and/or non-super-elastic characteristics and/or properties.
(143) In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy may be in the range of about 50 to about 60 weight percent nickel, with the remainder being essentially titanium. In some embodiments, the composition is in the range of about 54 to about 57 weight percent nickel. One example of a suitable nickel-titanium alloy is FHP-NT alloy commercially available from Furukawa Techno Material Co. of Kanagawa, Japan. Other suitable materials may include ULTANIUM™ (available from Neo-Metrics) and GUM METAL™ (available from Toyota). In some other embodiments, a superelastic alloy, for example a superelastic nitinol can be used to achieve desired properties. In at least some embodiments, portions or all of the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids a user in determining the location of the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. to achieve the same result.
(144) In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. For example, the medical device system **100**, the pusher wire **102**, the implant **104**, the sheath **106**, the locking element **108**, **200**, etc., and/or components or portions thereof, may be made of a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The medical device system **100**, the pusher wire **102**, the implant **104**, the sheath **106**, the locking element **108**, **200**, etc., or portions thereof, may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nitinol, and the like, and others.
(145) In some embodiments, the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, may be made from or include a polymer or other suitable material. Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane **85**A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name

PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), Marlex® high-density polyethylene, Marlex® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro(propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS **50**A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP). For example, the mixture can contain up to about 6 percent LCP.

(146) In some embodiments, the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. disclosed herein may include a fabric material disposed over or within the structure. The fabric material may be composed of a biocompatible material, such a polymeric material or biomaterial, adapted to promote tissue ingrowth. In some embodiments, the fabric material may include a bioabsorbable material. Some examples of suitable fabric materials include, but are not limited to, polyethylene glycol (PEG), nylon, polytetrafluoroethylene (PTFE, ePTFE), a polyolefinic material such as a polyethylene, a polypropylene, polyester, polyurethane, and/or blends or combinations thereof.

(147) In some embodiments, the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. may include and/or be formed from a textile material. Some examples of suitable textile materials may include synthetic yarns that may be flat, shaped, twisted, textured, pre-shrunk or un-shrunk. Synthetic biocompatible yarns suitable for use in the present invention include, but are not limited to, polyesters, including polyethylene terephthalate (PET) polyesters, polypropylenes, polyethylenes, polyurethanes, polyolefins, polyvinyl s, polymethylacetates, polyamides, naphthalene dicarboxylene derivatives, natural silk, and polytetrafluoroethylenes. Moreover, at least one of the synthetic yarns may be a metallic yarn or a glass or ceramic yarn or fiber. Useful metallic yarns include those yarns made from or containing stainless steel, platinum, gold, titanium, tantalum or a Ni—Co—Cr-based alloy. The yarns may further include carbon, glass or ceramic fibers. Desirably, the yarns are made from thermoplastic materials including, but not limited to, polyesters, polypropylenes, polyethylenes, polyurethanes, polynaphthalenes, polytetrafluoroethylenes, and the like. The yarns may be of the multifilament, monofilament, or spun-types. The type and denier of the yarn chosen may be selected in a manner which forms a biocompatible and implantable prosthesis and, more particularly, a vascular structure having desirable properties.

(148) In some embodiments, the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. may include and/or be treated with a suitable therapeutic agent. Some examples of suitable therapeutic agents may include anti-thrombogenic agents (such as heparin, heparin derivatives, urokinase, and PPack (dextrophenylalanine proline arginine chloromethylketone)); anti-proliferative agents (such as enoxaparin, angiopeptin, monoclonal antibodies capable of blocking smooth muscle cell proliferation, hirudin, and acetylsalicylic acid); anti-inflammatory agents (such as dexamethasone, prednisolone, corticosterone, budesonide, estrogen, sulfasalazine, and mesalamine); antineoplastic/antiproliferative/anti-mitotic agents (such as paclitaxel, 5-fluorouracil, cisplatin, vinblastine, vincristine, epothilones, endostatin, angiostatin and thymidine kinase inhibitors);

anesthetic agents (such as lidocaine, bupivacaine, and ropivacaine); anti-coagulants (such as D-Phe-Pro-Arg chloromethyl ketone, an RGD peptide-containing compound, heparin, anti-thrombin compounds, platelet receptor antagonists, anti-thrombin antibodies, anti-platelet receptor antibodies, aspirin, prostaglandin inhibitors, platelet inhibitors, and tick antiplatelet peptides); vascular cell growth promoters (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional activators, and translational promoters); vascular cell growth inhibitors (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional repressors, translational repressors, replication inhibitors, inhibitory antibodies, antibodies directed against growth factors, bifunctional molecules consisting of a growth factor and a cytotoxin, bifunctional molecules consisting of an antibody and a cytotoxin); cholesterol-lowering agents; vasodilating agents; and agents which interfere with endogenous vascoactive mechanisms.
(149) It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the invention. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A medical device system, comprising: a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath, the longitudinally extending slot terminating proximally of the intermediate region; a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an embolic coil positioned within the distal region of the lumen of the sheath; and a locking element configured to non-threadably engage the sheath, the locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the locking element, the lumen of the locking element having a generally hourglass shape; wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath; wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the sheath radially inwards onto a proximal end of the pusher wire.
2. The medical device system of claim 1, wherein the longitudinally extending slot of the sheath terminates at a circumferentially extending slit.
3. The medical device system of claim 1, wherein the sheath further comprises a first flap and a second flap adjacent the longitudinally extending slot.
4. The medical device system of claim 1, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.
5. The medical device system of claim 4, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.
6. The medical device system of claim 1, wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.
7. The medical device system of claim 1, wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.
8. A medical device system, comprising: a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending

from the proximal end to the distal end, the sheath further comprising a pair of longitudinally extending slots each extending distally from the proximal end of the sheath and forming a pair of flexible arms therebetween; a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an implant positioned within the lumen of the sheath; and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the locking element, the lumen of the locking element having a generally hourglass shape; wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath; wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the pair of flexible arms radially inwards against a proximal end of the pusher wire; wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.

9. The medical device system of claim 8, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.

10. The medical device system of claim 9, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.

11. The medical device system of claim 8, wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.

12. A medical device system, comprising: a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath; a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an embolic coil positioned within the lumen of the sheath; and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the locking element, the lumen of the locking element having a generally hourglass shape; wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath; and wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the sheath radially inwards against a proximal end of the pusher wire; wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.

13. The medical device system of claim 12, wherein the longitudinally extending slot of the sheath terminates at a circumferentially extending slit.

14. The medical device system of claim 12, wherein the sheath further comprises a first flap and a second flap adjacent the longitudinally extending slot.

15. The medical device system of claim 12, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.

16. The medical device system of claim 15, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.

17. The medical device system of claim 12, wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.

18. The medical device system of claim 12, wherein the locking element is configured to non-threadably engage the sheath.